Mohammad Tariqul Islam

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- Website: http://tariqul-islam.github.io
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- Github: https://github.com/tariqul-islam

Interest

Artificial Intelligence, Biomedical Engineering, Image Processing, Time Series

Education

2018 – 2024	**PhD. Electrical and Computer Engineering, Princeton University.**
2016 – 2018	**M.Sc. Electrical and Electronic Engineering, Bangladesh University of Engineering and Technology (BUET).**
2011 – 2016	**B.Sc. Electrical and Electronic Engineering, Bangladesh University of Engineering and Technology (BUET).**

Research Experience

2024 –	**MIT-Novo Nordisk Artificial Intelligence Fellow, MIT**, Supervised by, Deblina Sarker, AT&T Career Development Chair Professor, MIT
2019 – 2024	**Graduate Research Assistant, Imaging Physics Lab, Princeton**, Supervised by, Jason W. Fleischer, Professor, ECE, Princeton
2015 – 2018	**Masters Researcher, Computer Vision Group, BUET**, Supervised by, S.M. Mahbubur Rahman, Professor, EEE, BUET
2016 – 2018	**Computer Vision Researcher, Semion Inc.**, Supervised by, Khalid Ashraf, CEO, Semion
2014 – 2015	**Undergraduate Researcher, Signal Processing Group, BUET**, Supervised by, Shaikh Anowarul Fattah and Celia Shahnaz, Professor, EEE, BUET

Teaching Experience

2025	**Lecturer**, **MAS 746**: Next Generation Nano and Biotechnology, Bridging AI and Biotechnology
2021 – 2023	**Teaching Assistant**, **ECE 452**: Biomedical Imaging, **ECE 115**: Introduction to Computing - Programming Autonomous Vehicles.
2016 – 2018	**Lecturer**, BUET. **Courses**: Communication Theory, Circuit Theory, **Labs**: Digital Signal Processing, Numerical Techniques, Measurement and Instrumentation, Microwave Engineering.

Research Publications

Submitted

[1] Ahmed, K. M., **Islam**, **M. T.**, Zunaed, M., Jacob, M., & Hasan, T. (2026). *OrGAN: Towards organ-level image separation in projection radiographs using generative AI.*

Articles/Full Paper

[1] **Islam**, **M. T.** & Fleischer, J. W. (2025). The Shape of Attraction in UMAP: Exploring the Embedding Forces in Dimensionality Reduction. *arXiv preprint, arXiv:2503.09101*. , `https://bit.ly/islam2025shape`

[2] **Islam**, **M. T.**, Liu, D., & Sarkar, D. (2025b). Manifold Approximation leads to Robust Kernel Alignment. *arXiv preprint, arXiv:2510.22953*. , `https://bit.ly/islam2025kernel`

[3] **Islam**, **M. T.** & Fleischer, J. W. (2024a). Data Driven Phenotyping of COVID-19 using Chest X-rays. *preprint*. , `https://tariqul-islam.github.io/data/phenotype.pdf`

[4] **Islam**, **M. T.** & Fleischer, J. W. (2024b). Outlier Detection in Large Radiological Datasets Using UMAP. *International Workshop on Topology-and Graph-Informed Imaging Informatics*, 111–121. , `https://bit.ly/tariq2024outlier`

[5] Alam, M. M., **Islam**, **M. T.**, & Rahman, S. M. (2021). Unified Learning Approach for Egocentric Hand Gesture Recognition and Fingertip Detection. *Pattern Recognition*, 108200. , `https://bit.ly/alam2021unified`

[6] Saha, D., Rahman, S. M., **Islam**, **M. T.**, Ahmad, M. O., & Swamy, M. (2021). Prediction of Instantaneous Likeability of Advertisements using Deep Learning. *Cognitive Computation and Systems*. , `https://bit.ly/saha2021prediction`

[7] **Islam**, **M. T.** & Fleischer, J. W. (2020). Distinguishing L and H phenotypes of COVID-19 using a single x-ray image. *medRxiv*. , `https://bit.ly/islam2020dist`

[8] Alam, M. M. & **Islam**, **M. T.** (2019). Machine Learning Approach of Automatic Identification and Counting of Blood Cells. *Healthcare technology letters*, *6*(4), 103–108. , `https://bit.ly/alam2019cellid`

[9] Basnet, R., **Islam**, **M. T.**, Howlader, T., Rahman, S. M., & Hatzinakos, D. (2019). Estimation of Affective Dimensions using CNN-based Features of Audiovisual Data. *Pattern Recognition Letters*, *128*, 290–297. , `https://bit.ly/basnet2019estimation`

[10] **Islam**, **M. T.**, Ahmed, S. T., Shahnaz, C., & Fattah, S. A. (2019). SPECMAR: Fast Heart Rate Estimation from PPG Signal using a Modified Spectral Subtraction Scheme with Composite Motion Artifacts Reference Generation. *Medical & Biological Engineering & Computing*, *57*(3), 689–702. , `https://bit.ly/islam2018specmar`

[11] **Islam**, **M. T.**, Aowal, M. A., Minhaz, A. T., & Ashraf, K. (2018). Abnormality Detection and Localization in Chest X-Rays using Deep Convolutional Neural Networks. *arXiv preprint arXiv:1705.09850*. , `http://bit.ly/islam2017abnormality`

[12] **Islam**, **M. T.**, Rahman, S. M., Ahmad, M. O., & Swamy, M. (2018). Mixed Gaussian-Impulse Noise Reduction from Images Using Convolutional Neural Network. *Signal Processing: Image Communication*, *68*, 26–41. , `https://bit.ly/islam2018mixed`

[13] **Islam**, **M. T.**, Ahmed, S. T., Zabir, I., Shahnaz, C., & Fattah, S. A. (2017). Cascade and Parallel Combination (CPC) of Adaptive Filters for Estimating Heart Rate During Intensive Physical Exercise from Photoplethysmographic Signal. *Healthcare Technology Letters*. , `https://bit.ly/islam2017cpc`

[14] **Islam**, **M. T.**, Zabir, I., Ahamed, S. T., Yasar, M. T., Shahnaz, C., & Fattah, S. A. (2017). A Time-frequency Domain Approach of Heart Rate Estimation from Photoplethysmographic (PPG) Signal. *Biomedical Signal Processing and Control*, *36*, 146–154. , `https://bit.ly/islam2017timefreq`

Proceedings/Short Paper

[1] **Islam**, **M. T.**, Aarav, S., Wadood, S. A., & Fleischer, J. W. (2023). Super-resolution using Gradient Descent. In *Computational optical sensing and imaging, cw4d. 5.* , `https://bit.ly/tariq2023super`

[2] Fleischer, J. W. & **Islam**, **M. T.** (2020). Identifying and Phenotyping COVID-19 Patients using Machine Learning on Chest X-rays. In *European respiratory j.* (Vol. 56 (suppl 64)). European Respiratory J., 56 (suppl 64), Eur. Respiratory Soc. , `https://bit.ly/fleischer2020identifying`

[3] Das, S., Sikder, B., Khan, M. H. R., & **Islam**, **M. T.** (2018). Microstrip-Line Fed Gap-Coupled Antennas for Different Patch Geometries. In *2018 10th International Conference on Electrical and Computer Engineering (ICECE)* (pp. 473–476). IEEE. , `https://bit.ly/das2018microstrip`

[4] Hossain, M. S., Abir, M. T., Khan, M. H. R., & **Islam**, **M. T.** (2018). Multiheaded Starfish Shaped Multiband Microstrip Patch Antenna for Satellite Communication. In *2018 10th International Conference on Electrical and Computer Engineering (ICECE)* (pp. 449–452). IEEE. , `https://bit.ly/hossain2018multiheaded`

[5] **Islam**, **M. T.**, Saha, D., Rahman, S. M., Ahmad, M. O., & Swamy, M. (2018). A Variational Step for Reduction of Mixed Gaussian-Impulse Noise from Images. In *The International Conference on Electrical and Computer Engineering* (pp. 97–100). Dhaka, Bangladesh. , `https://bit.ly/islam2018variational`

[6] Najeeb, S., Sharmile, N., Khan, M. S., Sahin, I., **Islam**, **M. T.**, & Bhuiyan, M. I. H. (2018). Classification of Retinal Diseases from OCT scans using Convolutional Neural Networks. In *2018 10th International Conference on Electrical and Computer Engineering (ICECE)* (pp. 465–468). IEEE. , `https://bit.ly/najeeb2018classification`

[7] Basnet, R., **Islam**, **M. T.**, Howlader, T., Rahman, S., & Hatzinakos, D. (2017). Statistical Selection of CNN-Based Audiovisual Features for Instantaneous Estimation of Human Emotional States. In *The International Conference on New Trends in Computing Sciences* (pp. 50–54). Amman, Jordan. , `https://bit.ly/basnet2017stastical`

[8] Fattah, S. A., Rahman, M., Mustakin, N., **Islam**, **M. T.**, Khan, A. I., & Shahnaz, C. (2017). Wrist-Card: PPG Sensor based Wrist Wearable Unit for Low-Cost Personalized Cardio Healthcare System. In *IEEE Global Humanitarian Technology Conference* (pp. 1–7). San Jose, CA. , `https://bit.ly/fattah2017wrist`

[9] Hossain, M. M., Dipu, N. F., Islam, M. S., **Islam**, **M. T.**, Fattah, S. A., & Shahnaz, C. (2017). Design of a Low Cost Anti-Theft Sensor for Motorcycle Security Device. In *IEEE Region 10 Humanitarian Technology Conference* (pp. 778–783). Dhaka, Bangladesh. , `https://bit.ly/hossain2017motor`

[10] Nahian, M. A., Iftekhar, A., **Islam**, **M. T.**, Rahman, S., & Hatzinakos, D. (2017). CNN-Based Prediction of Frame-Level Shot Importance for Video Summarization. In *The International Conference on New Trends in Computing Sciences* (pp. 24–29). Amman, Jordan. , `https://bit.ly/nahian2017CNN`

[11] Ahamed, S. T. & **Islam**, **M. T.** (2016). An Efficient Method for Heart Rate Monitoring Using Wrist-type Photoplethysmographic Signals during Intensive Physical Exercise. In *Int. Conf.*

on Informatics, Electronics and Vision (pp. 863–868). IEEE. Dhaka,Bangladesh. , `https://bit.ly/ahmed2016ppg`

Non-archival Venues

[1] **Islam**, **M. T.**, Liu, D., & Sarkar, D. (2025a). *Kernel Alignment using Manifold Approximation.* 2025 ICLR Workshop on Representational Alignment (Re-Align). , `https://bit.ly/islam2025align`

[2] **Islam**, **M. T.** & Fleischer, J. W. (2022). *Manifold-aligned Neighbor Embedding.* 2022 ICLR Workshop on Geometrical and Topological Representation Learning. , `https://bit.ly/islam2022manifold`

Honors and Achievements

MIT-Novo Nordisk Artificial Intelligence Fellowship, at Massachusets Institute of Technology (MIT)

The William G. Bowen Merit Fellowship, at Princeton University

Dean's List Scholarship, In all levels of undergraduate studies at BUET

University Merit Scholarship, In all terms of undergraduate studies at BUET

Best Educational Impact, IEEE International Future Energy Challenge 2015, by IEEE Power and Energy Society

5th, IEEE Signal Processing Cup 2015, 5th among ~50 Teams, ICASSP-2015

Champion, International Robotics Challenge 2013-14, National Round, Dhaka, Bangladesh

Champion, Esonance 2015, MATLAB Challenge, Islamic University of Technology, Bangladesh

In News

2020 **AI tool gives doctors a new look at the lungs in treating COVID-19**, ECE, Princeton. `https://ece.princeton.edu/news/ai-tool-gives-doctors-new-look-lungs-treating-covid-19`

AI Can Pinpoint COVID-19 From Chest X-Rays, `https://www.medscape.com/viewarticle/937160`

Reviewer Experience

AISTATS 2026,

MICCAI 2025,

IEEE Transactions on Medical Imaging, IEEE

Medical Image Analysis, Elsevier

Biomedical Signal Processing and Control, Elsevier

Topology, Algebra, and Geometry in Machine Learning, 2022, Workshop at ICML 2022

2023 Topological, Algebraic, and Geometric Pattern Recognition with Applications Workshop, at CVPR 2023

IEEE Transactions on Services Computing, IEEE

IEEE Access, IEEE

Smart Health, Elsevier

Reviewer Experience (continued)

Circuits, Systems and Signal Processing, Springer
International Journal of Intelligent Robotics and Applications, Springer

Voluntary Experience

2018	**Founding Member and Secretary**, IEEE Signal Processing Society Bangladesh Chapter
	Secretary, IEEE EMBS Bangladesh Chapter
2017	**Organizing Committee Member**, 2017 2nd IEEE International Conference on Telecommunications and Photonics (ICTP), Dhaka, Bangladesh
	Co-Supervisor and Judge, EEE Day 2017 Robo War, BUET, Dhaka, Bangladesh
	Speaker, Training on Artificial Intelligence for Beginners, at Satyen Bose Science Club, BUET, Dhaka, Bangladesh
2016	**Technical Committee Member**, 2016 9th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh
	Organizer and Question Setter, EEE Day 2016 MATLAB Contest, BUET, Dhaka, Bangladesh
2015	**Organizer of Workshop**, Robot Tutor-1, IEEE BUET Student Branch, Dhaka, Bangladesh
	Webmaster, The Executive Committee of 2015, IEEE BUET Student Branch

Membership and Affiliation

2015 – Present	Institute of Electrical and Electronics Engineers (IEEE)
	IEEE Signal Processing Society